

2nd Youth International Math Olympiad
Division 2
Test A

Format: Questions are Integer Answers [-5,999].

Scoring: 1 pt each

Time: 80 minutes

1. Find the value of $x < 0$ such that $x^2 = 25$.

2. Let (a, b) be the point of intersection of

$$x = y + 2x + 1$$

and

$$x = y + 3x + 2$$

in the coordinate plane. Determine the value of $a + b$.

3. In $\triangle ABC$, D lies on side AC such that $BD = DC$. If $\angle ABC = 75^\circ$ and $\angle ACB = 40^\circ$, find the measure of $\angle ABD$.
4. How many 2 digit multiples of 8 are not factors of 64?
5. The probability it rains on any given day of the week is $\frac{1}{2}$. The probability that it rains on all but one of the days of the week can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n . Find $m + n$.

6. Let x be the unique solution to

$$8^x + 4^x = 4^{x+1} + 2^{x+1}.$$

The value of 4^{2x} can be expressed as $\frac{a+b\sqrt{c}}{d}$ where a, b, c and d are positive integers with $\gcd(a, b, d) = 1$ and c squarefree. Find $a + b + c + d$.

7. A $3 \times 3 \times 3$ cube is made up of 27 smaller $1 \times 1 \times 1$ glass blocks. A laser beam is fired through the cube. The beam travels in a perfectly straight line, entering through the center of a face of one of the 27 small blocks. The laser can only travel along lines that are parallel to the edges of the cube.

We wish to paint exactly 2 of the 27 small glass blocks black. How many different ways can we select the 2 blocks to paint such that it is impossible for a single laser beam to hit both black blocks at the same time?

8. Find the sum of all positive integers n such that at least $\frac{n}{2}$ of the integers between 1 and n divide into n .
9. Let ω_1 and ω_2 be circles of radius 5 and 12 respectively intersecting at X and Y . Denote the center of ω_1 as O_1 and the center of ω_2 as O_2 . If $O_1O_2 = 13$, the length of XY can be represented as $\frac{m}{n}$ where m and n are relatively prime. Find $m + n$.

10. If the roots of the polynomial $x^3 - 6x^2 + 11x - 7$ are a, b , and c , determine the value of

$$|(a + b - c)(b + c - a)(c + a - b)|.$$

11. Shining has 10 dice, one each of a fair i sided dice for all integers $6 \leq i \leq 15$. He rolls each of the dice once and adds the results. The probability that the sum of the values he gets has units digit 1 can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n . Find $m + n$.
12. How many perfect cubes less than 10^6 are not perfect squares?
13. Messi selects a random divisor of 2026^{2026} . The probability his chosen divisor is less than 2026^{1013} can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n . Find the last three digits of $m + n$.
14. Scaloni is holding tryouts for the Argentina World Cup team using 8 penalty-kick challenges. Each player must take part in exactly 3 of the 8 challenges. If no two players can share more than 1 challenge in common, determine the maximum number of players who can try out.
15. Find the number of ordered pairs of positive integers (m, n) such that $m \leq 2026$, $n \leq 2026$, and

$$\text{lcm}(m, n) = m + n + \text{gcd}(m, n) + 6.$$

16. Determine the last 3 digits of the value of

$$\sum_{i=1}^{2026} \left\lfloor \frac{i^2}{7} \right\rfloor = \left\lfloor \frac{1^2}{7} \right\rfloor + \left\lfloor \frac{2^2}{7} \right\rfloor + \cdots + \left\lfloor \frac{2026^2}{7} \right\rfloor.$$

17. Let $ABCD$ be a convex quadrilateral inscribed in a circle with $AB = 3$, $BC = 4$, $CD = 5$, and $DA = 6$. The value of $AC^2 + BD^2$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find the last three digits of $m + n$.
18. Let 11^a and 13^b be the highest powers of 11 and 13 that divide $2026!$ respectively. Find the remainder when

$$\frac{2026!}{11^a \cdot 13^b}$$

is divided by 143.

19. A football team needs to decide their next rosters by choosing from 2 already existing teams. Let the YIMO knights have 9 players and the NXT rabbits have 11 players, with the following rules for selections:
- The roster includes the same number of knights as rabbits.
 - There is at least one rabbit and knight on the roster.
 - One rabbit on the roster is named the rabbit captain and one knight on the roster is named the knight captain.

Two possible teams are considered distinct if it contains a different set of players or if either of the captains are different. Let N denote the number of ways the football team can be chosen. Find the last three digits of N .

20. An integer N is 123_b in base b and is 321_c in base c , where b and c are positive integers. Given that $b < 1000$, find the sum of all possible values of b .

2nd Youth International Math Olympiad

Division 2

Test B

Format: Questions are Integer Answers $[-5,999]$.

Scoring: 1 pt each

Time: 80 minutes

1. How many even positive integers are less than 2000?
2. At most how many regions can 3 circles divide the plane?
3. Lebron travels at a speed of 8 miles per hour and Jordan travels at a speed of 5 miles per hour. After traveling for 40 minutes, Lebron stops to wait for Jordan. How many minutes must he wait for Jordan to catch up?
4. Let $f(x)$ denote the sum of the digits of x . Determine the sum of all positive integers x for which $f(x) \geq x$.
5. Find the maximum positive integer value of x for which $x^2 - 8x + 12$ is a positive prime number.
6. If $x < 0$ and $((x + 1)^2 + 1)^2 + 1 = 290$, determine the last three digits of $((x - 1)^2 - 1)^2 - 1$.
7. Determine the number of diagonals of a regular octagon that do not pass through the center of the octagon.
8. A standard sheet of origami paper is a square with an area of 144. You fold the square perfectly in half to form a rectangle. Then, you fold that rectangle perfectly in half again to form a smaller square. Using a pair of scissors, you make a single straight cut that removes a corner of this folded square. When you completely unfold the paper, you find a hole exactly in the center of the sheet, and the perimeter of this hole is 24. What is the minimum possible area of the remaining paper?
9. Find the sum of all x such that

$$\sqrt{2x + 5} - \sqrt{x - 2} = 3.$$

10. Let N denote the number of 5 digit positive integers with their first 3 digits increasing and their last 3 digits decreasing. Find the last three digits of N .
11. Let x , y , and z be positive real numbers that satisfy the following system of equations:

$$x(y + z) = 125$$

$$y(z + x) = 89$$

$$z(x + y) = 164$$

Find the value of the product xyz .

12. Neymar must drop off 10 identical packages into 4 distinct mailboxes. Neymar distributes the packages so that each possible distribution of packages into mailboxes is equally likely. Once the packages are distributed, a mailbox is called overloaded if it contains 4 or more packages. Let E be the expected number of overloaded mailboxes. If E can be written as $\frac{m}{n}$, where m and n are relatively prime positive integers, find $m + n$.

13. How many real numbers $-\pi \leq x \leq \pi$ satisfy

$$\sin(10x) = 1 - \frac{x^2}{\pi^2}?$$

14. Shakira and Pitbull play a game starting with 200 coins on a table. On each turn, a player must remove at least one coin but cannot remove more than half of the coins currently on the table. A player loses if exactly one coin remains at the beginning of the turn. Shakira moves first. How many coins should she remove on her first turn to guarantee a win?

15. Find the number of positive integers $n \leq 1000$ such that the smallest prime dividing $\gcd(n, 2^n - 1)$ is equal to 7.

16. Positive integers a, b, c satisfy the property that

$$\log_a(b) + \log_b(c) = \log_a(c) = 2^k$$

for some positive integer k . The sum of all possible values of $\log_{ab}(c) - 1$ can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n . Find the value of $m + n$.

17. In an acute triangle ABC , the altitude from A meets BC at D . Let ω be the circle with diameter AD . Circle ω intersects the sides AB and AC again at E and F respectively. The tangent line to ω at E and the tangent line to ω at F intersect at a point P . Given that $AD = 24$, $BD = 10$, and $CD = 18$, the distance AP can be expressed in the form $\frac{a\sqrt{b}}{c}$ where a and c are relatively prime positive integers and b is squarefree. Find $a + b + c$.

18. Let x, y, z be positive real numbers that satisfy the following system of equations:

$$\begin{aligned}x^2 + xy + y^2 &= 40 \\y^2 + yz + z^2 &= 25 \\z^2 + zx + x^2 &= 49\end{aligned}$$

The value of $S = xy + yz + zx$ can be expressed as $a\sqrt{b}$ where a and b are positive integers with b squarefree. Find $a + b$.

19. Ronaldo is currently at the origin of the coordinate plane. In each move, he travels one unit either up, down, right, or left with equal probability. Let N denote the number of ways for Ronaldo to move 10 times such that he ends at the origin and doesn't pass through the origin before the final move. Find the last three digits of N .
20. Let N be a positive integer. When the last (rightmost) digit of N is moved to the very front, the resulting integer is exactly $\frac{13}{4}$ times the original number N . Let $N_1 < N_2 < N_3 < \dots$ be the infinite sequence of all such positive integers that satisfy this property. Find the last three digits in the decimal representation of N_{25} .

2nd Youth International Math Olympiad

Division 1

Test A

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1. In $\triangle ABC$, D lies on side AC such that $BD = DC$. If $\angle ABC = 75^\circ$ and $\angle ACB = 40^\circ$, find the measure of $\angle ABD$.

2. Let x be the unique solution to

$$8^x + 4^x = 4^{x+1} + 2^{x+1}.$$

The value of 4^{2x} can be expressed as $\frac{a+b\sqrt{c}}{d}$ where a, b, c and d are positive integers with $\gcd(a, b, d) = 1$ and c squarefree. Find $a + b + c + d$.

3. A $3 \times 3 \times 3$ cube is made up of 27 smaller $1 \times 1 \times 1$ glass blocks. A laser beam is fired through the cube. The beam travels in a perfectly straight line, entering through the center of a face of one of the 27 small blocks. The laser can only travel along lines that are parallel to the edges of the cube.

We wish to paint exactly 2 of the 27 small glass blocks black. How many different ways can we select the 2 blocks to paint such that it is impossible for a single laser beam to hit both black blocks at the same time?

4. Find the sum of all positive integers n such that at least $\frac{n}{2}$ of the integers between 1 and n divide into n .
5. Shining has 10 dice, one each of a fair i sided dice for all integers $6 \leq i \leq 15$. He rolls each of the dice once and adds the results. The probability that the sum of the values he gets has units digit 1 can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n . Find $m + n$.
6. How many perfect cubes less than 10^6 are not perfect squares?
7. Messi selects a random divisor of 2026^{2026} . The probability his chosen divisor is less than 2026^{1013} can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n . Find the last three digits of $m + n$.
8. Scaloni is holding tryouts for the Argentina World Cup team using 8 penalty-kick challenges. Each player must take part in exactly 3 of the 8 challenges. If no two players can share more than 1 challenge in common, determine the maximum number of players who can try out.
9. Find the number of ordered pairs of positive integers (m, n) such that $m \leq 2026$, $n \leq 2026$, and

$$\text{lcm}(m, n) = m + n + \gcd(m, n) + 6.$$

10. Determine the last 3 digits of the value of

$$\sum_{i=1}^{2026} \left\lfloor \frac{i^2}{7} \right\rfloor = \left\lfloor \frac{1^2}{7} \right\rfloor + \left\lfloor \frac{2^2}{7} \right\rfloor + \cdots + \left\lfloor \frac{2026^2}{7} \right\rfloor.$$

11. Consider $\triangle ABC$ with $AB = 4$, $BC = 3$, and $AC = 5$. Define point D on AB such that CD is the angle bisector of $\angle ACB$, and point E on BC such that AE is the angle bisector of $\angle BAC$. The distance from the incenter of $\triangle ABC$ to DE can be expressed as $\frac{\sqrt{a}}{b}$ for positive integers a and b with a squarefree. Find $a + b$.
12. Let 11^a and 13^b be the highest powers of 11 and 13 that divide $2026!$ respectively. Find the remainder when

$$\frac{2026!}{11^a \cdot 13^b}$$

is divided by 143.

13. A sequence of real numbers $a_1, a_2, a_3, \dots$ is defined by $a_1 = 2$ and the recurrence relation

$$a_{n+1} = \frac{a_n^4 + 6a_n^2 + 1}{4a_n^3 + 4a_n}$$

for all integers $n \geq 1$. The value of the infinite product

$$P = \prod_{n=1}^{\infty} \frac{4a_n(a_n^2 + 1)}{(a_n + 1)^3}$$

can be expressed in the form $\frac{m}{n}$, where m and n are relatively prime positive integers. Find the value of $m + n$.

14. Neil starts at the origin $(0, 0)$ on the coordinate plane. In a move, Neil moves one unit north, south, east or west, with the direction chosen randomly. Let (x, y) denote the final position of Neil after he makes 8 moves. Find the expected value of $x^4 y^2$.
15. A football team needs to decide their next rosters by choosing from 2 already existing teams. Let the YIMO knights have 9 players and the NXT rabbits have 11 players, with the following rules for selections:
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16. An integer N is 123_b in base b and is 321_c in base c , where b and c are positive integers. Given that $b < 1000$, find the sum of all possible values of b .

17. In an acute triangle $\triangle ABC$ with $BC = 13$, $AC = 14$ and $AB = 15$, the altitudes AD , BE , and CF intersect at the orthocenter H . A circle ω passes through A and H and is tangent to the circumcircle of $\triangle ABC$ at A . The circle ω intersects AB and AC at P and Q respectively. The length of PQ can be written as $\frac{m}{n}$ for relatively prime positive integers m and n . Find the last three digits of $m + n$.
18. Let $S = \{1, 2, 3, \dots, 10\}$. A non-empty subset $A \subseteq S$ is chosen uniformly at random from the set of all non-empty subsets of S . Let $x_1 < x_2 < \dots < x_k$ be the elements of A listed in strictly increasing order, where $k = |A|$ is the size of the subset.

We define the *weighted alternating sum* $S(A)$ of the subset to be

$$S(A) = \sum_{i=1}^k (-2)^{k-i} x_i = x_k - 2x_{k-1} + 4x_{k-2} - \dots + (-2)^{k-1} x_1$$

Let E be the expected value of $S(A)$. If $E = \frac{m}{n}$ for relatively prime positive integers m and n , find the last three digits of $m + n$.

19. Let $P(x)$ be a nonconstant polynomial with integer coefficients such that

$$P(x^2) = P(x)P(-x)$$

for all real x . If $P(1) \neq 0$ and $-2026 \leq P(2) \leq 2026$, how many possible polynomials could $P(x)$ be?

20. Let S be the set of all positive integers that are not a multiple of 3 between 1 and 100 inclusive. The sum of all possible values of

$$\sum_{k=1}^n \frac{1}{1 + 2 \cos \left(\frac{2k\pi}{n} \right)}$$

as n varies over the values of S can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n . Find $m + n$.

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2. A standard sheet of origami paper is a square with an area of 144. You fold the square perfectly in half to form a rectangle. Then, you fold that rectangle perfectly in half again to form a smaller square. Using a pair of scissors, you make a single straight cut that removes a corner of this folded square. When you completely unfold the paper, you find a hole exactly in the center of the sheet, and the perimeter of this hole is 24. What is the minimum possible area of the remaining paper?
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7. Let $f(x)$ denote the sum of the digits of x . Find the largest positive integer x for which $(f(x))^2 \geq x$.
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9. Shakira and Pitbull play a game starting with 200 coins on a table. On each turn, a player must remove at least one coin but cannot remove more than half of the coins currently on the table. A player loses if exactly one coin remains at the beginning of the turn. Shakira moves first. How many coins should she remove on her first turn to guarantee a win?
10. Let a , b , and c be the solutions to $x^3 - 3x^2 + 10x - 3 = 0$. Determine the value of

$$\frac{a^5}{b+c} + \frac{b^5}{a+c} + \frac{c^5}{a+b}.$$

11. Find the number of positive integers $n \leq 1000$ such that the smallest prime dividing $\gcd(n, 2^n - 1)$ is equal to 7.
12. Positive integers a, b, c satisfy the property that

$$\log_a(b) + \log_b(c) = \log_a(c) = 2^k$$

for some positive integer k . The sum of all possible values of $\log_{ab}(c) - 1$ can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n . Find the value of $m + n$.

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15. Ronaldo is currently at the origin of the coordinate plane. In each move, he travels one unit either up, down, right, or left with equal probability. Let N denote the number of ways for Ronaldo to move 10 times such that he ends at the origin and doesn't pass through the origin before the final move. Find the last three digits of N .
16. Let N be a positive integer. When the last (rightmost) digit of N is moved to the very front, the resulting integer is exactly $\frac{13}{4}$ times the original number N . Let $N_1 < N_2 < N_3 < \dots$ be the infinite sequence of all such positive integers that satisfy this property. Find the last three digits in the decimal representation of N_{25} .
17. In $\triangle ABC$, let I be the incenter and ω be the incircle. The circle ω is tangent to sides BC , CA , and AB at points D , E , and F , respectively. A line passing through I is perpendicular to AI and intersects the lines PE and PF at points X and Y , where P is the orthocenter of triangle DEF . Given that $BC = 20$, $AC = 21$, and $AB = 29$, the area of triangle PXY can be expressed in the form $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

- 18.** Let N be the number of subsets of $\{1, 2, 3, \dots, 16, 17\}$ which satisfy the property that the sum of their elements is a multiple of 5. Find the last three digits of N .
- 19.** Let the sequence of values E be defined by $E_1 = 2026$, and $E_m = 2026^{E_{m-1}}$ for all $m \geq 2$. Let the remainder when the sum

$$S = \sum_{k=1}^{4054^2} \gcd(k, 4054) \cdot k^{E_{2026}}$$

is divided by 4054^2 be X . Compute the last three digits of X .

- 20.** Let $a_1, a_2, \dots, a_5$ be real numbers satisfying

$$x^{10} + x^5 + 1 = \prod_{k=1}^5 (x^2 + a_k x + 1).$$

The value of

$$\sum_{k=1}^5 \frac{1}{a_k + 2}$$

can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n . Find $m + n$.